Ming Lin
Research Investigator
University of Michigan
linmin@umich.edu

2035C Palmer Commons Bldg
100 Washtenaw Avenue Ann Arbor, MI
48109-2218
http://minglin-home.github.io/

## Education

| | |
|---|---|
| 09/2004 - 08/2008 | Bachelor at Department of Automation, Tsinghua University, Beijing, China. |
| 09/2008 - 07/2014 | Ph.D. student at Department of Automation, Tsinghua University, Beijing, China.<br>o Advisor: Changshui Zhang.<br>o Innovate novel Nystrom sampling technique and its applications in manifold learning. |
| 11/2012 - 11/2013 | Visiting Research Scholar at Department of Computer Science and Engineering, Michigan State University, East Lansing, MI 48824, United States.<br>o Advisor: Rong Jin.<br>o Proposed several online random sampling learning methods to accelerate kernel machine without losing generalization performance. |
| 12/2013 - 7/2014 | Visiting Research Scholar at School of Computer Science, Carnegie Mellon University, Pittsburgh, PA 15213, United States.<br>o Advisor: Alexander G. Hauptmann.<br>o Developing multi-source fusion algorithms for video retrieval. |

## Experience

| | |
|---|---|
| 08/2014 - 08/2015 | Postdoctoral Research Fellow at School of Computer Science, Carnegie Mellon University, Pittsburgh, PA 15213, United States.<br>o Advisor: Alexander G. Hauptmann.<br>o Major developer of the ALADDIN project. |
| 09/2015 - Present | Postdoctoral Research Fellow at Department of Computational Medicine and Bioinformatics. University of Michigan Medical School, Ann Arbor, Michigan 48109, United States.<br>o Advisor: Jieping Ye.<br>o Proposed the second order linear model and applied it in various bioinformatic and computational medicine tasks. |

## Research Project

| | |
|---|---|
| 08/2017 - Present | **Large-scale Optimization on GPU** |

- Develop a scalable GPU computing platform for large-scale convex/nonconvex optimization
- Able to handle optimization problems that cannot be fitted into GPU on-board memory.
- Train models on very large datasets using one single GPU.
- Distributed to multiple GPUs

01/2015 - 12/2017 **The second order linear model: Theories and Applications.**

- Propose a novel model called the Second Order Linear Model (SLM)
- The SLM is able to capture high order feature interactions
- We applied SLM to Major Depressive Disorder (MDD) prediction and Alzheimer's Disease prediction.
- This project is a collaboration work with Janssen Research & Development, LCC.

07/2014 – 09/2015 **Automated Low-Level Analysis and Description of Diverse Intelligence Video (ALADDIN)**

- We developed a large-scale content-based video retrieval system with 47 features including 17 DCNN features.
- Our system ranked top-1 in 6 out of 8 tasks in MED 2014.
- This project is sponsored by Informedia@CMU.

## Research Interest

1. Large-scale non-convex optimization and statistical learning theory.
2. High-dimensional high-order statistics.
3. Random Matrices and tensor machines.
4. Real-world applications in computer vision, nature language processing and bioinformatics.

## Peer-Review Service

- Reviewer of International Joint Conference on Artificial Intelligence 2015, 2016, 2017.
- Senior Program Committee of IJCAI 2017.
- Reviewer of Journals:
  - IEEE Transactions on Pattern Analysis and Machine Intelligence.
  - ACM Transactions on Knowledge Discovery from Data.
  - Pattern Recognition
  - Neurocomputing.
  - Computer Vision and Image Understanding.
  - Data Mining and Knowledge Discovery

## Bibliography

*Peer-Reviewed Conferences*


1. Lijun Zhang, Jinfeng Yi, Ming Lin, Xiaofei He. Online Kernel Learning with a Near Optimal Sparsity Bound. In Proceedings of the 30th International Conference on Machine Learning (ICML), Pages 621 – 629, 2013.
2. Ming Lin, Rong Jin, Changshui Zhang. Efficient Sparse Recovery via Adaptive Non-Convex Regularizers with Oracle Property. In Proceedings of the 30th Conference on Uncertainty in Artificial Intelligenre (UAI), Pages 505-514, 2014.
3. Shoou-I Yu, Lu Jiang, Zexi Mao, Xiaojun Chang, Xingzhong Du, Chuang Gan, Zhenzhong Lan, Zhongwen Xu, Xuanchong Li, Yang Cai, Anurag Kumar, Yajie Miao, Lara Martin, Nikolas Wolfe, Shicheng Xu, Huan Li, Ming Lin, Zhigang Ma, Yi Yang, Deyu Meng, Shiguang Shan, Pinar Duygulu Sahin, Susanne Burger, Florian Metze, Rita Singh, Bhiksha Raj, Teruko Mitamura, Richard Stern, Alexander Hauptmann. Informedia@ TRECVID 2014 MED and MER. NIST TRECVID Video Retrieval Evaluation Workshop, 2014.
4. Ming Lin, Zhenzhong Lan, Alexander G. Hauptmann. Density Corrected Sparse Recovery when R.I.P. Condition is Broken. In Proceedings of the 24th International Joint Conference on Artificial Intelligence (IJCAI), Pages 3664-3670, 2015.
5. Zhenzhong Lan, Ming Lin, Xuanchong Li, Alexander G. Hauptmann, Bhiksha Raj. Beyond Gaussian Pyramid: Multi-skip Feature Stacking for Action Recognition. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Pages 204-212, 2015.
6. Chuang Gang, Ming Lin, Yi Yang, Alexander G. Hauptmann. Exploring Semantic Inter-Class Relationships (SIR) for Zero-Shot Action Recognition. In Proceedings of the 29th AAAI Conference on Artificial Intelligence (AAAI), Pages 3769-3775, 2015
7. Chuang Gang, Ming Lin, Yi Yang, Gerard de Melo, Alexander G. Hauptmann. Concepts Not Alone: Exploring Pairwise Relationships for Zero-Shot Video Activity Recognition. In Proceedings of the 30th AAAI Conference on Artificial Intelligence (AAAI), Pages 3487-3493, 2016.
8. Ming Lin, Jieping Ye. A Non-convex One-Pass Framework for Generalized Factorization Machine and Rank-One Matrix Sensing. In Proceedings of the 30th Annual Conference on Neural Information Processing Systems (NIPS), Pages 1633-1641, 2016.
9. Xiang Li, Aoxiao Zhong, Ming Lin, Ning Guo, Mu Sun, Arkadiusz Sitek, Jieping Ye, James Thrall, Quanzheng Li. Self-paced Convolutional Neural Network for Computer Aided Detection in Medical Imaging Analysis. In Proceedings of the 8th International Workshop on Machine Learning in Medical Imaging (MLMI), 2017.
10. Tieliang Gong, Guangtao Wang, Jieping Ye, Zongben Xu, Ming Lin. Margin Based PU Learning. The Thirty-Second AAAI Conference on Artificial Intelligence (AAAI), 2018.

*Peer-Reviewed Journals*

1. Shizhun Yang, Ming Lin, Chenping Hou, Changshui Zhang, Yi Wu. A General Framework for Transfer Sparse Subspace Learning. Neural Computing and Applications. Volume 21, Number 7, Pages 1801-1817, August 2012.
2. Ming Lin, Shifeng Weng, Changshui Zhang. On the Sample Complexity of Random Fourier Features for Online Learning. ACM Transactions on Knowledge Discovery from Data (TKDD), Volume 8 Issue 3, Pages 13:1--13:19, June 2014.
3. Ming Lin, Fei Wang, Changshui Zhang. Large-Scale Eigenvector Approximation via Hilbert Space Embedding Nystrom. Pattern Recognition (PR), 48(5), Pages 1904-1912, 2015.
4. Zheng Pan, Ming Lin, Guangdong Hou, Changshui Zhang. Damping proximal coordinate descent algorithm for non-convex regularization. Neurocomputing, vol 152, Pages 151-163, 2015.
5. Zhen Hu, Ming Lin, Changshui Zhang. Dependent Online Kernel Learning with Constant Number of Random Fourier Features. IEEE Transactions on Neural Networks and Learning Systems (TNNLS), Volume 26, Issue 10, Pages 2464-2476, 2015.
6. Ming Lin, Lijun Zhang, Rong Jin, Shifeng Weng, Changshui Zhang. Online Kernel Learning with Nearly Constant Support Vectors. Neurocomputing. Volume 179, Pages 26-36, 2016.
7. Shoou-I Yu, Yi Yang, Zhongwen Xu, Shicheng Xu, Deyu Meng, Zexi Mao, Zhigang Ma, Ming Lin, Xuanchong Li, Huan Li, Zhenzhong Lan, Lu Jiang, Alexander G. Hauptmann, Chuang Gan, Xingzhong Du, Xiaojun Chang. Strategies for Searching Video Content with Text Queries or Video Examples (Invited Paper). ITE Transactions on Media Technology and Applications 4.3, Pages 227-238, 2016
8. Xiaojun Chang, Zhigang Ma, Ming Lin, Yi Yang, Alexander G. Hauptmann. Feature Interaction Augmented Sparse Learning for Fast Kinect Motion Detection. IEEE Transactions on Image Processing, Volume 26, Issue 8, Pages 3911-3920. 2017.
9. Daqing Chang, Ming Lin, Changshui Zhang. On the generalization ability of online gradient descent algorithm under the quadratic growth condition. To appear in TNNLS 2018.